

JAWAD CHAUDHARY

MOST FREQUENTLY REPEATED CONCEPTUAL AREAS IN BT302 IMMUNOLOGY CONCEPT CRACKER MCQS FINAL EXAMS

Main Group (BS Biological Sciences) *https://chat.whatsapp.com/DWXwJpXahbxFVQutey8JXN*

BT302 – Immunology *https://chat.whatsapp.com/HJJISsZ75dzDZXHe1CK1Wz*

Q1: The process by which B cells produce antibodies against an immunogen is called antibody _____.
Answer: Formation

Q2: Antibody formation begins when a B cell recognizes an antigen through its _____.
Answer: B-cell receptor (BCR)

Q3: Activated B cells differentiate into antibody-secreting _____.
Answer: Plasma cells

Q4: Plasma cells are specialized for the secretion of _____.
Answer: Antibodies

Q5: The ability of the immune system to distinguish self from non-self is called self/non-self _____.
Answer: Discrimination

Q6: Immune destruction of self-tissues by antibodies is known as _____.
Answer: Autoimmunity

Q7: The enhanced immune response upon second exposure to the same antigen is called the _____ response.
Answer: Memory

Q8: Another name for the memory immune response is the _____ response.
Answer: Anamnestic

Q9: Memory against previously encountered antigens is mainly maintained by _____ cells.
Answer: Memory B

Q10: Antibodies react specifically with the antigen that induced their _____.
Answer: Production

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Q11: High antigen specificity minimizes unnecessary _____ reactions.
Answer: Cross

Q12: Antibody specificity is a characteristic feature of _____ immunity.
Answer: Adaptive

Q13: The first exposure of the body to an immunogen is called the _____ injection.
Answer: Primary

Q14: Re-exposure to the same immunogen is known as the _____ injection.
Answer: Secondary

Q15: The first phase of immunogen clearance is the _____ phase.
Answer: Equilibrium

Q16: During the equilibrium phase, immunogen diffuses between vascular and _____ compartments.
Answer: Extravascular

Q17: The second phase of immunogen clearance is the _____ phase.
Answer: Catabolic

Q18: Macrophages actively participate in the _____ phase of immunogen clearance.
Answer: Catabolic

Q19: Phagocytic cells are mainly responsible for antigen elimination during the _____ phase.
Answer: Catabolic

Q20: Newly synthesized antibodies combine with antigens to form immune _____.
Answer: Complexes

Q21: Immune complexes are mainly removed by _____.
Answer: Phagocytosis

Q22: Antibodies become detectable in serum after the _____ elimination phase.
Answer: Immune

Q23: During secondary exposure, pre-existing circulating antibodies cause rapid antigen _____.
Answer: Elimination

Q24: The major antibody produced during the primary immune response is _____.
Answer: IgM

Q25: Antibodies produced during the primary response generally have low _____.
Answer: Affinity

Q26: The major antibody produced during the secondary immune response is _____.
Answer: IgG

Q27: Secondary immune responses may also produce increased levels of IgA and _____.
Answer: IgE

Q28: Antibodies produced during the secondary response are more _____ for antigen binding.
Answer: Affinitive

Q29: Increased antibody affinity after repeated antigen exposure is called affinity _____.
Answer: Maturation

Q30: Lower doses of antigen favor increased antibody _____.
Answer: Affinity

Q31: Antibody responses against protein antigens are generally _____-dependent.
Answer: T-cell

Q32: The first phase of the primary response to T-dependent antigens is the _____ phase.
Answer: Lag

Q33: During the lag phase, lymphocytes undergo proliferation and _____.
Answer: Differentiation

Q34: The lag phase of the primary T-dependent response usually lasts about _____ days.
Answer: 5–7

Q35: The rapid rise in antibody concentration occurs during the _____ phase.
Answer: Log

Q36: The log phase is characterized by rapid activation of _____ cells.
Answer: B

Q37: Plasma cells are generated mainly during the _____ phase.
Answer: Log

Q38: Antibody concentration remains relatively constant during the _____ phase.
Answer: Plateau

Q39: During the plateau phase, antibody synthesis equals antibody _____.
Answer: Decay

Q40: Falling antibody levels are observed during the _____ phase.
Answer: Decline

Q41: During the secondary response, the lag phase is _____ than in the primary response.
Answer: Shorter

Q42: The rapidity of the secondary response is mainly due to immunological _____.
Answer: Memory

Q43: The log phase of the secondary response produces _____ antibody levels than the primary response.
Answer: Higher

Q44: The plateau phase is generally absent during the _____ response to T-dependent antigens.
Answer: Secondary

Q45: Antibodies produced during the secondary response may persist for months or even _____.
Answer: Years

Q46: T-independent antigens predominantly stimulate production of _____ antibodies.
Answer: IgM

Q47: T-independent antigens generally fail to produce immunological _____.
Answer: Memory

Q48: Secondary responses against T-independent antigens lack a significant _____ response.
Answer: Memory

Q49: During the primary response, the earliest antibody secreted by plasma cells is _____.
Answer: IgM

Q50: Antibody class switching during the secondary response mainly occurs from IgM to _____.
Answer: IgG

Q51: The pool of memory cells is mainly composed of activated _____ and B lymphocytes.
Answer: T

Q52: Some plasma cells differentiate into _____ cells after antigen exposure.
Answer: Memory

Q53: During the secondary response, antibody class switching mainly results in increased _____ production.
Answer: IgG

Q54: The earliest cellular event during the primary response is antigen recognition by _____ receptors.
Answer: B-cell

Q55: B-cell activation results in clonal expansion and cellular _____.
Answer: Differentiation

Q56: Plasma cells originate from activated _____ lymphocytes.
Answer: B

Q57: Antibody secretion reaches its maximum during the _____ phase.
Answer: Log

Q58: Death of plasma cells contributes to the _____ phase of antibody response.
Answer: Decline

Q59: T-cell activity decreases as antigen becomes _____.
Answer: Depleted

Q60: During the plateau phase, newly synthesized antibodies are balanced by antibody _____.
Answer: Decay

Q61: Immunoglobulin genes undergo somatic _____ during B-cell development.
Answer: Recombination

Q62: Antibody diversity is primarily generated by V(D)J _____.
Answer: Recombination

Q63: The inherited form of immunoglobulin gene segments is called the _____ configuration.
Answer: Germline

Q64: Immunoglobulin genes are divided into _____ and heavy chain families.
Answer: Light

Q65: Human light chains are classified as kappa and _____.
Answer: Lambda

Q66: The lambda light-chain gene is located on chromosome _____.
Answer: 22

Q67: The kappa light-chain gene is located on chromosome _____.
Answer: 2

Q68: Heavy-chain genes are located on chromosome _____.
Answer: 14

Q69: Lambda light-chain genes contain approximately _____ variable (V) gene segments.
Answer: 30

Q70: Kappa light-chain genes contain approximately _____ variable (V) gene segments.
Answer: 250

Q71: The joining segment in light-chain genes is represented by the letter _____.
Answer: J

Q72: The variable region of the heavy chain is formed by V, D, and _____ segments.
Answer: J

Q73: Diversity (D) gene segments are present only in the _____ chain genes.
Answer: Heavy

Q74: Non-coding sequences separating exons are called _____.
Answer: Introns

Q75: The leader exon encodes the _____ peptide.
Answer: Signal

Q76: The constant region determines the _____ of an immunoglobulin.
Answer: Class

Q77: The variable region is primarily responsible for antigen _____.
Answer: Binding

Q78: During B-cell maturation, only one V segment is joined with one _____ segment in light chains.
Answer: J

Q79: Rearrangement of immunoglobulin genes occurs before antigen _____.
Answer: Exposure

Q80: Somatic recombination occurs during _____-cell development.
Answer: B

Q81: V(D)J recombination produces enormous antibody _____.
Answer: Diversity

Q82: Gene rearrangement occurs only in developing _____ cells.
Answer: B

Q83: The recombination process is mediated by RAG-1 and _____ proteins.
Answer: RAG-2

Q84: Recombination signal sequences guide V(D)J _____.
Answer: Recombination

Q85: Junctional diversity results from random addition or deletion of _____.
Answer: Nucleotides

Q86: Terminal deoxynucleotidyl transferase is abbreviated as _____.
Answer: TdT

Q87: TdT increases antibody diversity by adding random _____.
Answer: Nucleotides

Q88: Combinational diversity results from different heavy and _____ chain pairings.
Answer: Light

Q89: The antigen-binding site is formed by both heavy and _____ chains.
Answer: Light

Q90: The complementarity-determining regions are abbreviated as _____.
Answer: CDRs

Q91: CDRs are located within the _____ regions of immunoglobulins.
Answer: Variable

Q92: Hypervariable regions directly interact with the _____.
Answer: Antigen

Q93: Constant regions show much less sequence variation than _____ regions.
Answer: Variable

Q94: Somatic hypermutation occurs after antigen _____.
Answer: Stimulation

Q95: Somatic hypermutation contributes to affinity _____.
Answer: Maturation

Q96: Affinity maturation increases the strength of antigen-_____ interaction.
Answer: Antibody

Q97: Class switching changes the constant region while preserving antigen _____.
Answer: Specificity

Q98: Class switching occurs after activation of mature _____ cells.
Answer: B

Q99: During class switching, the variable region remains _____.
Answer: Unchanged

Q100: Expression of immunoglobulin genes ultimately results in production of functional _____.
Answer: Antibodies

Q101: The DNA rearrangement that joins V, D, and J segments occurs only in developing _____ cells.
Answer: B

Q102: Expression of immunoglobulin genes begins after successful gene _____.
Answer: Rearrangement

Q103: Only one heavy-chain allele is expressed in a mature B cell due to _____ exclusion.
Answer: Allelic

Q104: Allelic exclusion ensures that a B cell produces antibodies of a single _____.
Answer: Specificity

Q105: Productive rearrangement of one allele suppresses rearrangement of the other _____.
Answer: Allele

Q106: Rearrangement of the heavy-chain gene occurs before the _____-chain gene.
Answer: Light

Q107: Expression of μ (mu) chain marks the formation of the _____-B cell receptor.
Answer: Pre

Q108: The Pre-B-cell receptor is expressed before complete _____-chain rearrangement.
Answer: Light

Q109: Successful light-chain rearrangement leads to expression of surface _____.
Answer: IgM

Q110: Mature naïve B cells co-express surface IgM and surface _____.
Answer: IgD

Q111: IgM and IgD expressed on naïve B cells possess identical antigen _____.
Answer: Specificity

Q112: Co-expression of IgM and IgD occurs through alternative RNA _____.
Answer: Splicing

Q113: Alternative RNA processing allows production of different immunoglobulin _____.
Answer: Isotypes

Q114: Secreted antibodies differ from membrane-bound antibodies mainly at the _____ region.
Answer: Carboxyl-terminal

Q115: Membrane immunoglobulins contain a hydrophobic _____ domain.
Answer: Transmembrane

Q116: Secreted antibodies lack the _____ region present in membrane immunoglobulins.
Answer: Transmembrane

Q117: Class switching changes antibody _____ without altering antigen specificity.
Answer: Isotype

Q118: During class switching, the V(D)J region remains _____.
Answer: Unchanged

Q119: Heavy-chain class switching occurs in the _____ region of the gene.
Answer: Constant

Q120: Switch recombination takes place at specialized _____ regions.
Answer: Switch

Q121: The enzyme primarily responsible for class switching is _____.
Answer: Activation-induced cytidine deaminase (AID)

Q122: The same enzyme involved in class switching also participates in somatic _____.
Answer: Hypermutation

Q123: Somatic hypermutation mainly affects the _____ region of immunoglobulin genes.
Answer: Variable

Q124: High-affinity B-cell clones are selected during affinity _____.
Answer: Maturation

Q125: Affinity maturation mainly occurs within the _____ centers of lymphoid follicles.
Answer: Germinal

Q126: B cells with higher-affinity receptors receive stronger _____ signals.
Answer: Survival

Q127: Low-affinity B cells are eliminated mainly through _____.
Answer: Apoptosis

Q128: Memory B cells are generated primarily after _____ stimulation.
Answer: Antigen

Q129: Plasma cells are specialized mainly for antibody _____.
Answer: Secretion

Q130: Memory B cells persist for long periods because of immunological _____.
Answer: Memory

Q131: The principal function of memory B cells is to produce a rapid _____ response.
Answer: Secondary

Q132: During the secondary response, plasma cells produce antibodies with higher _____.
Answer: Affinity

Q133: Class switching allows antibodies to perform different _____ functions.
Answer: Effector

Q134: IgG is the predominant antibody produced after class _____.
Answer: Switching

Q135: Antibody diversity is increased by random pairing of heavy and _____ chains.
Answer: Light

Q136: Junctional diversity is generated during V(D)J gene _____.
Answer: Joining

Q137: Nucleotide deletion during recombination contributes to antibody _____.
Answer: Diversity

Q138: Addition of non-templated nucleotides occurs through the action of _____.
Answer: TdT

Q139: Recombination Signal Sequences are commonly abbreviated as _____.
Answer: RSS

Q140: RSS guide the accurate joining of immunoglobulin gene _____.
Answer: Segments

Q141: The 12/23 rule ensures proper V(D)J _____.
Answer: Recombination

Q142: The 12/23 rule prevents incorrect joining of gene _____.
Answer: Segments

Q143: Functional immunoglobulin genes are produced only after successful _____.
Answer: Recombination

Q144: Failure of productive gene rearrangement may lead to B-cell _____.
Answer: Death

Q145: Mature plasma cells contain extensive rough endoplasmic _____.
Answer: Reticulum

Q146: Plasma cells possess a prominent _____ apparatus for antibody secretion.
Answer: Golgi

Q147: Activated B cells undergo rapid _____ expansion after antigen recognition.
Answer: Clonal

Q148: Clonal selection results in expansion of antigen-specific _____ cells.
Answer: B

Q149: Each plasma cell secretes antibodies with identical antigen _____.
Answer: Specificity

Q150: The remarkable diversity of antibodies enables the immune system to recognize an almost unlimited number of _____.
Answer: Antigens

Q151: The branch of immunology that studies antigen-antibody reactions in vitro is called _____.
Answer: Serology

Q152: The visible interaction between antigen and antibody is known as a _____ reaction.
Answer: Serological

Q153: Antigen-antibody reactions are highly _____ in nature.
Answer: Specific

Q154: The strength with which a single antigenic determinant binds an antibody is called _____.
Answer: Affinity

Q155: The overall binding strength between multivalent antigen and antibody is called _____.
Answer: Avidity

Q156: Precipitation reactions occur when the antigen is in _____ form.
Answer: Soluble

Q157: Agglutination reactions occur when the antigen is in _____ form.
Answer: Particulate

Q158: Maximum precipitation occurs at the _____ zone.
Answer: Equivalence

Q159: Excess antibody prevents lattice formation in the _____ zone.
Answer: Prozone

Q160: Excess antigen prevents visible precipitation in the _____ zone.
Answer: Postzone

Q161: Lattice formation requires antigens and antibodies to be _____.
Answer: Multivalent

Q162: Radial immunodiffusion is mainly used for _____ quantification.
Answer: Antigen

Q163: Double immunodiffusion is also known as the _____ technique.
Answer: Ouchterlony

Q164: Immunoelectrophoresis combines electrophoresis with _____.
Answer: Immunodiffusion

Q165: Rocket immunoelectrophoresis is mainly used for quantitative _____ estimation.
Answer: Antigen

Q166: Complement fixation test detects the presence of specific _____.
Answer: Antibodies

Q167: In complement fixation test, absence of hemolysis indicates a _____ result.
Answer: Positive

Q168: Hemolysis in complement fixation test indicates a _____ result.
Answer: Negative

Q169: Sheep red blood cells are commonly used as _____ cells in complement fixation tests.
Answer: Indicator

Q170: Hemolysin is an antibody directed against _____ erythrocytes.
Answer: Sheep

Q171: Neutralization reactions prevent the biological activity of toxins or _____.
Answer: Viruses

Q172: Toxin neutralization depends upon specific _____ molecules.
Answer: Antibodies

Q173: Opsonization enhances phagocytosis by coating microorganisms with _____.
Answer: Antibodies

Q174: IgG is the principal immunoglobulin involved in _____.
Answer: Opsonization

Q175: ELISA stands for Enzyme-Linked _____ Assay.
Answer: Immunosorbent

Q176: The enzyme commonly used in ELISA is linked to an _____.
Answer: Antibody

Q177: Direct ELISA is primarily used to detect _____.
Answer: Antigens

Q178: Indirect ELISA is mainly used to detect _____.
Answer: Antibodies

Q179: Sandwich ELISA requires _____ antibodies against the same antigen.
Answer: Two

Q180: Competitive ELISA is particularly useful for detecting _____ antigens.
Answer: Small

Q181: Color development in ELISA occurs after addition of a suitable _____.
Answer: Substrate

Q182: The intensity of color in ELISA is proportional to the amount of _____.
Answer: Antigen

Q183: Western blotting identifies proteins after separation by gel _____.
Answer: Electrophoresis

Q184: In Western blotting, proteins are transferred onto a _____ membrane.
Answer: Nitrocellulose

Q185: Immunofluorescence uses antibodies labeled with _____ dyes.
Answer: Fluorescent

Q186: Flow cytometry analyzes cells based on their size, granularity, and _____.
Answer: Fluorescence

Q187: Fluorescent antibodies are commonly used in _____ cytometry.
Answer: Flow

Q188: Monoclonal antibodies are produced from a single clone of _____ cells.
Answer: B

Q189: Hybridoma technology combines antibody-producing B cells with _____ cells.
Answer: Myeloma

Q190: Hybridoma cells are both immortal and antibody-_____ cells.
Answer: Producing

Q191: Polyclonal antibodies recognize _____ epitopes on the same antigen.
Answer: Multiple

Q192: Monoclonal antibodies recognize only _____ epitope.
Answer: One

Q193: Diagnostic immunology commonly employs monoclonal antibodies because of their high _____.
Answer: Specificity

Q194: Pregnancy test kits commonly utilize the _____ principle.
Answer: Immunochromatography

Q195: Rapid diagnostic kits are based on antigen-antibody _____.
Answer: Reactions

Q196: Agglutination tests are commonly used for blood _____.
Answer: Grouping

Q197: Latex agglutination utilizes antigen-coated _____ particles.
Answer: Latex

Q198: Immunological assays depend primarily upon antigen-antibody _____.
Answer: Specificity

Q199: Most immunodiagnostic techniques are designed to detect either antigens or _____.
Answer: Antibodies

Q200: The foundation of all serological tests is the highly specific interaction between antigen and _____.
Answer: Antibody

Q201: Hypersensitivity reactions are immune responses that cause _____ to host tissues.
Answer: Damage

Q202: Immediate hypersensitivity is also known as Type _____ hypersensitivity.
Answer: I

Q203: Type I hypersensitivity is primarily mediated by _____ antibodies.
Answer: IgE

Q204: Mast cells possess high-affinity receptors for _____.
Answer: IgE

Q205: Cross-linking of IgE on mast cells results in _____.
Answer: Degranulation

Q206: The major vasoactive amine released from mast cells is _____.
Answer: Histamine

Q207: Histamine primarily increases vascular _____.
Answer: Permeability

Q208: The first exposure to an allergen usually results in _____.
Answer: Sensitization

Q209: Allergic rhinitis is an example of Type _____ hypersensitivity.
Answer: I

Q210: Systemic anaphylaxis is a severe form of Type _____ hypersensitivity.
Answer: I

Q211: Type II hypersensitivity is mediated mainly by IgG and _____ antibodies.
Answer: IgM

Q212: In Type II hypersensitivity, antibodies react against antigens present on the _____ surface.
Answer: Cell

Q213: Complement activation plays an important role in Type _____ hypersensitivity.
Answer: II

Q214: Hemolytic disease of the newborn is a classic example of Type _____ hypersensitivity.
Answer: II

Q215: Blood transfusion incompatibility mainly involves Type _____ hypersensitivity.
Answer: II

Q216: Type III hypersensitivity results from deposition of immune _____.
Answer: Complexes

Q217: Immune complexes are commonly deposited in blood vessel _____.
Answer: Walls

Q218: Serum sickness is an example of Type _____ hypersensitivity.
Answer: III

Q219: The Arthus reaction is a localized Type _____ hypersensitivity reaction.
Answer: III

Q220: Complement activation contributes significantly to tissue injury in Type _____ hypersensitivity.
Answer: III

Q221: Delayed-type hypersensitivity is classified as Type _____ hypersensitivity.
Answer: IV

Q222: Type IV hypersensitivity is primarily mediated by _____ lymphocytes.
Answer: T

Q223: Type IV hypersensitivity usually develops within _____ hours after antigen exposure.
Answer: 24–72

Q224: Tuberculin skin testing is an example of Type _____ hypersensitivity.
Answer: IV

Q225: Contact dermatitis is primarily mediated by Type _____ hypersensitivity.
Answer: IV

Q226: Activated macrophages are important effector cells in Type _____ hypersensitivity.
Answer: IV

Q227: Autoimmune diseases develop when immune tolerance to _____ antigens is lost.
Answer: Self

Q228: The immune system normally avoids attacking the body's own tissues through self-_____.
Answer: Tolerance

Q229: Autoantibodies are antibodies directed against _____ antigens.
Answer: Self

Q230: Rheumatoid arthritis is considered an _____ disease.
Answer: Autoimmune

Q231: Systemic lupus erythematosus is commonly abbreviated as _____.
Answer: SLE

Q232: Immunodeficiency disorders result from impaired _____ function.
Answer: Immune

Q233: Primary immunodeficiencies are generally caused by _____ defects.
Answer: Genetic

Q234: Secondary immunodeficiencies are usually acquired during an individual's _____.
Answer: Lifetime

Q235: AIDS is caused by infection with the _____ virus.
Answer: HIV

Q236: HIV primarily infects _____ T lymphocytes.
Answer: CD4+

Q237: Progressive loss of CD4+ T cells leads to severe _____ deficiency.
Answer: Immunity

Q238: Opportunistic infections are common in patients with severe _____ deficiency.
Answer: Immune

Q239: Vaccination induces _____ immunity.
Answer: Active

Q240: Administration of preformed antibodies provides _____ immunity.
Answer: Passive

Q241: Live attenuated vaccines contain organisms with reduced _____.
Answer: Virulence

Q242: Killed vaccines contain microorganisms that are no longer _____.
Answer: Viable

Q243: Booster doses primarily stimulate _____ immune responses.
Answer: Secondary

Q244: Vaccine-induced protection largely depends on immunological _____.
Answer: Memory

Q245: Adjuvants are added to vaccines to enhance the _____ response.
Answer: Immune

Q246: Herd immunity reduces disease transmission within a _____.
Answer: Population

Q247: Passive immunity provides immediate but _____ protection.
Answer: Temporary

Q248: Active immunity generally develops more _____ than passive immunity.
Answer: Slowly

Q249: Successful vaccination results in the formation of long-lived _____ cells.
Answer: Memory

Q250: The principal goal of immunization is to establish protective immunological _____.
Answer: Memory

Q251: A tissue graft exchanged between genetically identical individuals is called an _____ graft.
Answer: Isograft

Q252: A graft transferred between two genetically different individuals of the same species is known as an _____.
Answer: Allograft

Q253: A graft obtained from an individual of another species is called a _____.
Answer: Xenograft

Q254: Successful organ transplantation depends largely on compatibility of _____ molecules.
Answer: MHC

Q255: Human MHC molecules are collectively known as the _____ system.
Answer: HLA

Q256: HLA stands for Human _____ Antigen.
Answer: Leukocyte

Q257: Class I MHC molecules are expressed on nearly all _____ cells.
Answer: Nucleated

Q258: Class II MHC molecules are primarily expressed on professional antigen-presenting _____.
Answer: Cells

Q259: The strongest transplantation antigens are encoded by the _____ complex.
Answer: MHC

Q260: Graft rejection is mainly mediated by _____ lymphocytes.
Answer: T

Q261: Hyperacute graft rejection occurs due to pre-existing _____.
Answer: Antibodies

Q262: Acute graft rejection usually develops within _____ after transplantation.
Answer: Days

Q263: Chronic graft rejection develops over a period of _____.
Answer: Months

Q264: Immunosuppressive drugs are administered to reduce graft _____.
Answer: Rejection

Q265: Cyclosporine primarily suppresses activation of _____ lymphocytes.
Answer: T

Q266: Corticosteroids reduce graft rejection by suppressing the _____ response.
Answer: Immune

Q267: Bone marrow transplantation is commonly used to treat disorders of the _____ system.
Answer: Hematopoietic

Q268: Graft-versus-host disease mainly occurs after _____ marrow transplantation.
Answer: Bone

Q269: In graft-versus-host disease, donor _____ cells attack recipient tissues.
Answer: T

Q270: Matching HLA antigens increases the probability of graft _____.
Answer: Survival

Q271: Cytokines function primarily as chemical _____ between immune cells.
Answer: Messengers

Q272: Cytokines exert their effects by binding to specific cell-surface _____.
Answer: Receptors

Q273: Interleukins are mainly involved in communication between _____ cells.
Answer: Immune

Q274: Interferons play an important role in defense against _____ infections.
Answer: Viral

Q275: Tumor necrosis factor is commonly abbreviated as _____.
Answer: TNF

Q276: Colony-stimulating factors regulate the production of blood _____.
Answer: Cells

Q277: Chemokines primarily direct the _____ of leukocytes.
Answer: Migration

Q278: Cytokines may act in autocrine, paracrine, or _____ manners.
Answer: Endocrine

Q279: Excessive cytokine release may result in a cytokine _____.
Answer: Storm

Q280: Interleukin-2 is essential for _____-cell proliferation.
Answer: T

Q281: Immunological tolerance prevents immune responses against _____ antigens.
Answer: Self

Q282: Central tolerance is established during lymphocyte _____.
Answer: Development

Q283: Peripheral tolerance controls self-reactive lymphocytes outside the _____ organs.
Answer: Primary

Q284: Failure of immune tolerance predisposes an individual to _____ diseases.
Answer: Autoimmune

Q285: Clonal deletion eliminates strongly self-reactive _____.
Answer: Lymphocytes

Q286: Regulatory T cells help maintain immune _____.
Answer: Tolerance

Q287: Regulatory T cells suppress excessive immune _____.
Answer: Responses

Q288: Immunological memory is a characteristic feature of the _____ immune system.
Answer: Adaptive

Q289: Secondary immune responses are both faster and more _____ than primary responses.
Answer: Stronger

Q290: Memory lymphocytes are generated after successful antigen _____.
Answer: Exposure

Q291: Humoral immunity is primarily mediated by _____.
Answer: Antibodies

Q292: Cell-mediated immunity is primarily mediated by _____ lymphocytes.
Answer: T

Q293: Plasma cells are terminally differentiated _____ lymphocytes.
Answer: B

Q294: Memory B cells rapidly differentiate into _____ cells following re-exposure to antigen.
Answer: Plasma

Q295: Antibody affinity generally increases after repeated antigen _____.
Answer: Exposure

Q296: Class switching allows antibodies to retain antigen specificity while changing their _____ function.
Answer: Effector

Q297: Most protective vaccines depend upon induction of long-lived _____ cells.
Answer: Memory

Q298: The effectiveness of adaptive immunity depends on antigen _____.
Answer: Specificity

Q299: The hallmark of the adaptive immune response is specificity and immunological _____.
Answer: Memory

Q300: The ultimate objective of the adaptive immune system is to eliminate pathogens while protecting _____ tissues.
Answer: Host

🚨 Concept Cracker MCQS Most Important Repeated Conceptual 🚨

⚠ Important Note:

These MCQs have been compiled from repeated exam patterns, important concepts, and previous assessments to support your preparation.

📚 However, do NOT rely solely on this compilation.

For the best results, thoroughly study your Handouts, PPTs, Video Lectures, and Course Material as well.

Prepare Smart, Work Hard, and Aim for Excellence.

Best of Luck! 🍀📚

BS Biological Sciences Team 🧬🎓

JAWAD CHAUDHARY